

From Absolute Zero to Score Propagators: Foundations from Complex Numbers to Diffusion-Model Research on Dynamic Objects

Scientific note (self-contained)

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Prologue: the through-line and why these topics belong together

We want a *single mental model* in which the same core structures reappear as:

- **complex phases** in quantum mechanics,
- **connections/curvature** in geometry and gauge theory,
- **generators/propagators** in deterministic and stochastic dynamics,
- **scores** (gradients of log-densities) in statistical inference and modern diffusion generative models,
- **operators** (Koopman, transfer/Perron–Frobenius) that propagate observables and densities,
- **learned propagators** that aim to compress and identify underlying physical dynamics.

The research endpoint is:

Central research problem (preview). Given a dynamic object with internal (physical) time τ (e.g. a trajectory of a system), and a diffusion model with diffusion time t (noise level), we can define a family of *score fields*

$$s_t^\tau(x) := \nabla_x \log p_t^\tau(x),$$

where p_t^τ is the (possibly marginal) density of the state x at internal time τ under diffusion level t . The goal is to understand *what it means* for s_t^τ to evolve with τ , *why a propagator should exist*, and *how learning such a propagator relates to learning the underlying physical dynamics*.

We will build everything needed to make this statement fully precise.

1 Part I: Mathematical and Physical Foundations

1.1 Real numbers: completeness as a stability principle

Intuition. Physics and probability repeatedly require taking limits: refining a partition in an integral, taking a time step $\Delta t \rightarrow 0$, or defining Brownian motion as a limit of random walks. A number system (or space of functions) must be *closed under these limits* to avoid “falling out” of the space.

Definition (real numbers and completeness). The real numbers $\mathbb{R}$ can be characterized (among other equivalent ways) as a *complete ordered field*: it is a field (supports $+$, $\times$ and inverses), it is ordered, and it is *complete* in the sense that every Cauchy sequence converges in $\mathbb{R}$.

Definition (Cauchy sequence). Let (X, d) be a metric space, where $d : X \times X \rightarrow [0, \infty)$ is a distance. A sequence $(x_n)_{n \in \mathbb{N}}$ is *Cauchy* if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall m, n \geq N : d(x_m, x_n) < \varepsilon.$$

Intuitively: the sequence becomes arbitrarily self-consistent. Convergence implies Cauchy, but the converse requires completeness.

Definition (completeness of a metric space). A metric space (X, d) is *complete* if every Cauchy sequence in X converges to a limit in X .

Why completeness matters for stochastic processes. The Itô integral is defined as an L^2 -limit of Riemann-sum-like approximations (Section 5.3). To guarantee that the limit exists *within the space*, we rely on the completeness of the relevant function spaces (Hilbert spaces like L^2) and, at a foundational level, on complete probability spaces (Section 1.6).

1.2 Complex numbers: why $\mathbb{C}$ is the minimal extension that enables phases

Motivation. On $\mathbb{R}$, the equation $x^2 + 1 = 0$ has no solution because $x^2 \geq 0$. Complex numbers are the smallest extension of $\mathbb{R}$ that contains a solution.

Definition (complex numbers). A complex number is an ordered pair $(a, b) \in \mathbb{R}^2$ written as

$$z = a + ib,$$

where i is defined by $i^2 = -1$. The *real part* is $\Re(z) = a$, the *imaginary part* is $\Im(z) = b$.

Geometry of $\mathbb{C}$. We identify $z = a + ib$ with the point (a, b) in the plane. Addition corresponds to vector addition. Multiplication has a geometric meaning after introducing polar coordinates.

Modulus and conjugation. The complex conjugate is $\bar{z} = a - ib$. The modulus (Euclidean length) is

$$|z| := \sqrt{a^2 + b^2}.$$

One checks directly that $z\bar{z} = |z|^2$.

1.3 Power series and Euler's formula: why rotations are exponentials

Power series. A *power series* centered at $x_0 \in \mathbb{R}$ is an “infinite polynomial”

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n,$$

where $(a_n)_{n \geq 0} \subset \mathbb{R}$ (or $\mathbb{C}$). It converges for $|x - x_0| < R$ for some radius $R \in [0, \infty]$.

Defining e^x , $\sin x$, $\cos x$ by power series. For $x \in \mathbb{R}$ (and in fact $x \in \mathbb{C}$), define

$$e^x := \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \cos x := \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \quad \sin x := \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}.$$

These converge for all $x \in \mathbb{C}$.

Euler's formula. Plug $x = i\theta$ into the exponential series:

$$e^{i\theta} = \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} = \sum_{n=0}^{\infty} \frac{(i\theta)^{2n}}{(2n)!} + \sum_{n=0}^{\infty} \frac{(i\theta)^{2n+1}}{(2n+1)!}.$$

Since $i^{2n} = (-1)^n$ and $i^{2n+1} = i(-1)^n$,

$$e^{i\theta} = \sum_{n=0}^{\infty} (-1)^n \frac{\theta^{2n}}{(2n)!} + i \sum_{n=0}^{\infty} (-1)^n \frac{\theta^{2n+1}}{(2n+1)!} = \cos \theta + i \sin \theta.$$

This is Euler's formula.

Polar form and rotation. Every nonzero $z \in \mathbb{C} \setminus \{0\}$ can be written

$$z = re^{i\theta},$$

where $r = |z| > 0$ and $\theta = \arg(z)$ is an angle (defined modulo 2π). Multiplying by $e^{i\theta}$ rotates vectors in the plane: complex multiplication unifies scaling and rotation.

1.4 Vector spaces, norms, inner products: linear structure as the language of superposition

Vector space. A (real or complex) vector space V is a set equipped with addition and scalar multiplication satisfying the usual axioms (associativity, distributivity, etc.). Physics uses vector spaces because linear combinations encode *superposition* (Part IV).

Norm. A norm $\|\cdot\| : V \rightarrow [0, \infty)$ satisfies: (i) $\|v\| = 0 \Leftrightarrow v = 0$, (ii) $\|\alpha v\| = |\alpha| \|v\|$, (iii) $\|u + v\| \leq \|u\| + \|v\|$.

Inner product. An inner product on a complex vector space V is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$ such that

$$\langle u, v \rangle = \overline{\langle v, u \rangle}, \quad \langle u, u \rangle \geq 0, \quad \langle u, u \rangle = 0 \Leftrightarrow u = 0, \quad \langle u, \alpha v + \beta w \rangle = \alpha \langle u, v \rangle + \beta \langle u, w \rangle.$$

It induces a norm $\|v\| := \sqrt{\langle v, v \rangle}$.

1.5 Metric spaces: distance as a minimal notion of geometry

Metric space. A metric space (X, d) consists of a set X and a function $d : X \times X \rightarrow [0, \infty)$ such that: (i) $d(x, y) = 0 \Leftrightarrow x = y$, (ii) $d(x, y) = d(y, x)$, (iii) triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$.

Completeness revisited. Completeness of (X, d) is about limit stability (Section 1.1). In stochastic analysis, we repeatedly use complete metric spaces of functions (e.g. continuous paths with uniform norm; square-integrable random variables with L^2 norm).

1.6 Probability measures: measure theory as the substrate of randomness

Why measure theory appears. Probability must handle uncountably many outcomes (e.g. continuous random variables), limits, and conditioning. Measure theory provides a consistent way to define “sizes” of sets and integrals of functions.

Definition (measurable space). A measurable space is a pair $(\Omega, \mathcal{F})$ where Ω is a set and $\mathcal{F}$ is a σ -algebra (a collection of subsets containing Ω , closed under complements and countable unions).

Definition (probability measure). A probability measure is a function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ such that $\mathbb{P}(\Omega) = 1$ and $\mathbb{P}$ is countably additive:

$$A_i \in \mathcal{F} \text{ disjoint} \Rightarrow \mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

A probability space is $(\Omega, \mathcal{F}, \mathbb{P})$.

Random variable. A random variable is a measurable function $X : \Omega \rightarrow \mathbb{R}^d$ (for some dimension $d \in \mathbb{N}$). “Measurable” ensures probabilities of events like $\{X \in A\}$ are defined for measurable sets A .

Expectation. If X is integrable, the expectation is

$$\mathbb{E}[X] := \int_{\Omega} X(\omega) \, d\mathbb{P}(\omega).$$

If X has a density p on $\mathbb{R}^d$ (with respect to Lebesgue measure dx), then

$$\mathbb{E}[f(X)] = \int_{\mathbb{R}^d} f(x)p(x) \, dx.$$

Variance and covariance. For scalar X :

$$\text{Var}(X) := \mathbb{E}[(X - \mathbb{E}[X])^2].$$

For vector $X \in \mathbb{R}^d$:

$$\text{Cov}(X) := \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^\top] \in \mathbb{R}^{d \times d}.$$

2 Part II: Classical Mechanics $\rightarrow$ Statistical Physics

2.1 Coordinates and frames: labels versus physics

Coordinates. A coordinate system is a rule that assigns d real numbers to each point in a region of space (more generally, of a manifold; Part III). Coordinates are *not physical*; they are descriptive.

Cartesian coordinates in $\mathbb{R}^2$. A point is $(x, y) \in \mathbb{R}^2$.

Polar coordinates in $\mathbb{R}^2$. A point is (r, θ) with $r \geq 0$ and $\theta \in [0, 2\pi)$, related by

$$x = r \cos \theta, \quad y = r \sin \theta.$$

The *metric* (infinitesimal distance) becomes

$$ds^2 = dr^2 + r^2 d\theta^2.$$

This formula already encodes geometry: angular motion costs distance proportional to radius r .

2.2 Inertial versus non-inertial frames: why fictitious forces appear

Inertial frame. A frame is inertial if Newton’s laws hold without additional “fictitious” forces. In an inertial frame, a free particle moves at constant velocity.

Non-inertial frame. A frame accelerating or rotating relative to an inertial frame is non-inertial. Newton’s second law can still be written, but additional terms appear that depend on the frame.

2.3 Centrifugal, Coriolis, and Euler forces

Setup and notation. Let $\mathbf{r}(t) \in \mathbb{R}^3$ be the position of a particle in a rotating frame with angular velocity $\boldsymbol{\Omega}(t) \in \mathbb{R}^3$. Let $\mathbf{v}$ be velocity in the rotating frame. Let m be the particle’s mass.

Centrifugal force (rotation, even if $\dot{\Omega} = 0$)

Formula. In a rotating frame, there appears an outward force

$$\mathbf{F}_{\text{cent}} = -m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}).$$

Why this form. The double cross product extracts the component of $\mathbf{r}$ perpendicular to the rotation axis, producing magnitude $m\Omega^2 r_{\perp}$ pointing outward.

Coriolis force (motion *within* a rotating frame)

Formula.

$$\mathbf{F}_{\text{Cor}} = -2m\boldsymbol{\Omega} \times \mathbf{v}.$$

Key properties. It is perpendicular to $\mathbf{v}$, so it does no work:

$$\mathbf{F}_{\text{Cor}} \cdot \mathbf{v} = 0.$$

Thus it changes direction but not speed.

Euler force (time-dependent rotation rate)

Formula. If $\dot{\Omega}(t) \neq 0$, an additional force appears:

$$\mathbf{F}_{\text{Eul}} = -m\dot{\boldsymbol{\Omega}} \times \mathbf{r}.$$

Physical meaning. It arises when the rotating frame is *speeding up or slowing down*.

2.4 Polar-coordinate derivation of “centrifugal” terms in $\mathbb{R}^2$

Why derive it explicitly. This makes tangible that centrifugal-like terms come from *time-dependent basis vectors* (geometry of coordinates).

Kinematics in polar basis. In $\mathbb{R}^2$, define orthonormal unit vectors $\hat{\mathbf{e}}_r(\theta)$ and $\hat{\mathbf{e}}_{\theta}(\theta)$:

$$\hat{\mathbf{e}}_r = (\cos \theta, \sin \theta), \quad \hat{\mathbf{e}}_{\theta} = (-\sin \theta, \cos \theta).$$

Position is $\mathbf{r} = r\hat{\mathbf{e}}_r$. Differentiate:

$$\dot{\hat{\mathbf{e}}}_r = \dot{\theta}\hat{\mathbf{e}}_{\theta}, \quad \dot{\hat{\mathbf{e}}}_{\theta} = -\dot{\theta}\hat{\mathbf{e}}_r.$$

Velocity:

$$\dot{\mathbf{r}} = \dot{r}\hat{\mathbf{e}}_r + r\dot{\theta}\hat{\mathbf{e}}_{\theta}.$$

Acceleration:

$$\ddot{\mathbf{r}} = (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{e}}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\mathbf{e}}_{\theta}.$$

The radial term $-r\dot{\theta}^2$ is the centrifugal acceleration in polar coordinates.

2.5 Potential energy, force fields, gradients, and curl

Scalar potential and conservative forces. A *potential energy* is a scalar function $V : \mathbb{R}^d \rightarrow \mathbb{R}$ such that the force field $\mathbf{F} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ satisfies

$$\mathbf{F}(\mathbf{x}) = -\nabla_{\mathbf{x}} V(\mathbf{x}),$$

where $\nabla_{\mathbf{x}}$ is the gradient with respect to position $\mathbf{x} \in \mathbb{R}^d$.

Meaning of the gradient. For $V : \mathbb{R}^d \rightarrow \mathbb{R}$, ∇V is the vector of partial derivatives:

$$\nabla V(\mathbf{x}) := \left(\frac{\partial V}{\partial x_1}, \dots, \frac{\partial V}{\partial x_d} \right).$$

It points in the direction of steepest increase of V ; hence $-\nabla V$ points toward steepest decrease.

Work and line integrals. The work done by a force along a curve C parameterized by $\mathbf{r}(s)$ is

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_{s=a}^b \mathbf{F}(\mathbf{r}(s)) \cdot \mathbf{r}'(s) ds.$$

If $\mathbf{F} = -\nabla V$ (conservative), then $W = V(\mathbf{r}(a)) - V(\mathbf{r}(b))$ depends only on endpoints.

Curl and circulation in $\mathbb{R}^3$. For a vector field $\mathbf{F} = (F_1, F_2, F_3)$, define the curl

$$\nabla \times \mathbf{F} := \left(\frac{\partial F_3}{\partial x_2} - \frac{\partial F_2}{\partial x_3}, \frac{\partial F_1}{\partial x_3} - \frac{\partial F_3}{\partial x_1}, \frac{\partial F_2}{\partial x_1} - \frac{\partial F_1}{\partial x_2} \right).$$

Stokes' theorem relates circulation to curl:

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S},$$

where S is an oriented surface, ∂S its boundary loop, and $d\mathbf{S}$ its oriented area element. If $\mathbf{F} = -\nabla V$, then $\nabla \times \mathbf{F} = 0$ (locally curl-free).

2.6 Pressure, temperature, entropy, chemical potential: why thermodynamics is geometry of constraints

Thermodynamic state variables. A macroscopic state can often be described by *extensive* variables (scale with system size): internal energy U , entropy S , volume V , particle number N . Conjugate *intensive* variables do not scale with size: temperature T , pressure P , chemical potential μ .

First law (differential form). For a simple compressible system,

$$dU = T dS - P dV + \mu dN.$$

Each term is “intensive $\times$ extensive”. This form *defines* the conjugate variables:

$$T := \left(\frac{\partial U}{\partial S} \right)_{V,N}, \quad P := - \left(\frac{\partial U}{\partial V} \right)_{S,N}, \quad \mu := \left(\frac{\partial U}{\partial N} \right)_{S,V}.$$

These are *response coefficients*: they quantify how energy changes under infinitesimal changes of the extensive variables.

Pressure as momentum flux. Microscopically, pressure is force per unit area, $P = F/A$, and can be interpreted as momentum transfer rate to a boundary per unit area. Its units are energy per volume: $[P] = \text{J}/\text{m}^3$.

Entropy as multiplicity. In statistical mechanics, if a macrostate corresponds to Ω microstates, entropy is

$$S = k_B \log \Omega,$$

where k_B is Boltzmann's constant. For a probability distribution p over microstates x , a more general form is Shannon/Gibbs entropy:

$$S[p] = -k_B \int p(x) \log p(x) \, dx.$$

Chemical potential. μ is the energetic cost of adding a particle at fixed (S, V) :

$$\mu = \left(\frac{\partial U}{\partial N} \right)_{S, V}.$$

It is the driving potential for diffusion and reactions: particle flow tends to reduce gradients in μ .

2.7 Legendre duality: changing which variables you control

Why transform. Experiments often control intensive variables like T and P rather than S and V . Legendre transforms systematically switch between these descriptions without changing physics.

Definition (Legendre transform in one dimension). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex and differentiable. Define the conjugate variable $p = f'(x)$. The Legendre transform is

$$f^*(p) := \sup_{x \in \mathbb{R}} (px - f(x)).$$

At the maximizer $x(p)$, one has $p = f'(x)$ and $f^*(p) = p x(p) - f(x(p))$.

Thermodynamic potentials. Starting from $U(S, V, N)$:

$$F(T, V, N) := U - TS \quad (\text{Helmholtz free energy}),$$

where $T = (\partial U / \partial S)_{V, N}$. Also

$$H(S, P, N) := U + PV \quad (\text{enthalpy}), \quad G(T, P, N) := U - TS + PV \quad (\text{Gibbs free energy}).$$

These are Legendre transforms of U with respect to selected variables.

2.8 Wind as pressure-gradient-driven flow: a quantitative sketch

Pressure-gradient acceleration. In fluid mechanics, for air density ρ and pressure field $P(\mathbf{x})$, the pressure-gradient force per unit mass is

$$\mathbf{a}_{\text{pg}} = -\frac{1}{\rho} \nabla P.$$

This accelerates air from high to low pressure.

Coriolis balance (geostrophic wind). On Earth, the Coriolis acceleration is $-2\mathbf{\Omega} \times \mathbf{v}$, where $\mathbf{\Omega}$ is Earth's rotation vector. At midlatitudes, a dominant balance in large-scale flows is

$$-2\mathbf{\Omega} \times \mathbf{v} \approx -\frac{1}{\rho} \nabla P,$$

which implies wind flows roughly along isobars (not directly down the gradient). The typical Coriolis parameter is $f = 2\Omega \sin \phi$ (latitude ϕ), giving a scale

$$v \sim \frac{1}{\rho f} \frac{\Delta P}{L},$$

where ΔP is pressure change across length L . This is a quantitative bridge between thermodynamic fields and inertial geometry.

3 Part III: Geometry and Symmetry

3.1 Manifolds and coordinate charts: local Euclidean structure

Intuition. Many spaces look like $\mathbb{R}^d$ *locally* but not globally (sphere, torus). Physics needs this to describe curved spaces (spacetime, configuration spaces, probability manifolds).

Definition (smooth manifold). A d -dimensional smooth manifold M is a set equipped with an atlas of coordinate charts $(U_\alpha, \varphi_\alpha)$ where $U_\alpha \subset M$ are open sets covering M and $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^d$ are homeomorphisms such that transition maps $\varphi_\beta \circ \varphi_\alpha^{-1}$ are smooth on overlaps.

Fields on manifolds. A *scalar field* is a function $f : M \rightarrow \mathbb{R}$. A *vector field* assigns to each point $p \in M$ a tangent vector in $T_p M$.

3.2 Tangent spaces: what vectors at a point" really mean

Definition (tangent space). At a point $p \in M$, the tangent space $T_p M$ is a d -dimensional vector space that represents directions of curves through p . One equivalent definition: tangent vectors are derivations on smooth functions at p .

Coordinates. If $(x^1, \dots, x^d)$ are local coordinates, then $\{\partial/\partial x^i\}_{i=1}^d$ is a basis of $T_p M$.

3.3 Riemannian geometry: metric, geodesics, connection, curvature

Metric tensor

Definition (Riemannian metric). A Riemannian metric on M is a smoothly varying inner product $g_p(\cdot, \cdot)$ on each tangent space $T_p M$. In coordinates, it is represented by a symmetric positive-definite matrix $g_{ij}(x)$ such that the squared infinitesimal length is

$$ds^2 = \sum_{i,j=1}^d g_{ij}(x) dx^i dx^j.$$

Why a metric is needed. The metric defines lengths, angles, volumes, and therefore kinetic energy in mechanics and distances on statistical manifolds.

Covariant derivative and Christoffel symbols

Why ordinary derivatives fail. Vectors at different points belong to different tangent spaces; subtraction is not canonical. We need a rule to compare them.

Definition (connection / covariant derivative). A (linear) connection ∇ assigns to vector fields X, Y a new vector field $\nabla_X Y$ satisfying linearity in X , Leibniz rule in Y with respect to scalar multiplication, and tensoriality in X .

Levi-Civita connection. On a Riemannian manifold (M, g) there is a unique connection that is: (i) metric-compatible ($\nabla g = 0$), (ii) torsion-free (defined below). Its coefficients in coordinates are the Christoffel symbols

$$\Gamma_{ij}^k = \frac{1}{2} \sum_{\ell=1}^d g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}),$$

where $\partial_i := \partial/\partial x^i$ and $(g^{k\ell})$ is the inverse matrix of $(g_{k\ell})$.

Covariant derivative of a vector field. If $v = v^k \partial_k$ is a vector field, then

$$(\nabla_i v)^k := \partial_i v^k + \Gamma_{ij}^k v^j,$$

where repeated indices are summed (Einstein convention).

Geodesics

Definition (geodesic). A curve $\gamma(t)$ is a geodesic if its velocity vector is parallel transported along itself:

$$\nabla_{\dot{\gamma}(t)} \dot{\gamma}(t) = 0.$$

In coordinates $x^k(t)$ this becomes

$$\ddot{x}^k(t) + \Gamma_{ij}^k(x(t)) \dot{x}^i(t) \dot{x}^j(t) = 0.$$

Geodesics generalize straight lines to curved spaces.

Parallel transport and holonomy

Parallel transport. Given a curve $\gamma(t)$ and a vector field $v(t) \in T_{\gamma(t)} M$ along it, v is *parallel transported* if

$$\nabla_{\dot{\gamma}(t)} v(t) = 0.$$

This defines a linear map between tangent spaces at endpoints.

Holonomy. For a closed loop C based at p , parallel transport around C yields a linear transformation $H(C) : T_p M \rightarrow T_p M$. The set of all such transformations forms the *holonomy group*. For small loops, the deviation from identity is controlled by curvature.

Curvature and torsion

Riemann curvature tensor. Curvature measures the non-commutativity of covariant derivatives. For a vector field v ,

$$(\nabla_i \nabla_j - \nabla_j \nabla_i)v^k = R_{\ell ij}^k v^\ell,$$

defining the components $R_{\ell ij}^k$.

Ricci tensor and scalar curvature. Contract indices:

$$\text{Ric}_{ij} := R_{ikj}^k, \quad R := g^{ij} \text{Ric}_{ij}.$$

In general relativity, Ricci curvature enters Einstein's equations; in thermodynamic geometry, curvature often signals interaction or criticality.

Torsion. Torsion measures the failure of infinitesimal parallelograms to close. For vector fields X, Y ,

$$T(X, Y) := \nabla_X Y - \nabla_Y X - [X, Y],$$

where $[X, Y]$ is the Lie bracket (commutator) of vector fields. In coordinates, torsion components are

$$T_{ij}^k = \Gamma_{ij}^k - \Gamma_{ji}^k.$$

The Levi-Civita connection is torsion-free: $T = 0$.

3.4 Lie groups and Lie algebras: symmetry as smooth transformation

Lie group. A Lie group G is a smooth manifold with a group operation $(g, h) \mapsto gh$ and inversion $g \mapsto g^{-1}$ that are smooth maps.

Examples. $U(1) = \{e^{i\theta} : \theta \in \mathbb{R}\}$ (complex phases), $SO(3)$ (rotations in $\mathbb{R}^3$), $SU(2)$ (spin rotations), $SU(3)$ (color symmetry in QCD).

Lie algebra. The Lie algebra $\mathfrak{g}$ of G is the tangent space at the identity $e \in G$, equipped with a bilinear antisymmetric bracket $[\cdot, \cdot]$ capturing infinitesimal commutators. Infinitesimal generators $X \in \mathfrak{g}$ exponentiate to group elements $e^{\varepsilon X}$ for small ε .

Infinitesimal generators (preview for dynamics). The “generator” concept will reappear in Part V (Markov semigroups) and Part IV (Hamiltonian generating unitary evolution). The common pattern is:

$$\text{finite evolution} = \exp(\text{time} \times \text{generator}).$$

4 Part IV: Quantum and Field-Theoretic Language

4.1 Hilbert spaces: why quantum states are vectors

Hilbert space. A Hilbert space $\mathcal{H}$ is a complex vector space with an inner product $\langle \cdot, \cdot \rangle$ such that $\mathcal{H}$ is complete under the induced norm $\|\psi\| = \sqrt{\langle \psi, \psi \rangle}$.

Example (wavefunctions). For a particle in $\mathbb{R}^3$, a standard choice is $\mathcal{H} = L^2(\mathbb{R}^3)$, the space of square-integrable complex functions $\psi : \mathbb{R}^3 \rightarrow \mathbb{C}$ with inner product

$$\langle \phi, \psi \rangle = \int_{\mathbb{R}^3} \overline{\phi(\mathbf{x})} \psi(\mathbf{x}) d^3x.$$

Born rule. If $\psi(\mathbf{x}, t)$ is a normalized wavefunction ($\int |\psi|^2 = 1$), then $|\psi(\mathbf{x}, t)|^2$ is a probability density over positions.

4.2 Bra–ket notation: coordinates as inner products

States. A quantum state is a unit vector $|\psi\rangle \in \mathcal{H}$, defined up to a global phase $|\psi\rangle \sim e^{i\alpha}|\psi\rangle$.

Position eigenkets. $|\mathbf{x}\rangle$ denotes the (generalized) eigenstate of position at $\mathbf{x} \in \mathbb{R}^3$. The position-space wavefunction is the *coordinate* of $|\psi\rangle$ in the position basis:

$$\psi(\mathbf{x}) = \langle \mathbf{x} | \psi \rangle.$$

Completeness (resolution of identity). Formally,

$$\int_{\mathbb{R}^3} |\mathbf{x}\rangle \langle \mathbf{x}| d^3x = \text{Id},$$

so

$$|\psi\rangle = \int_{\mathbb{R}^3} |\mathbf{x}\rangle \psi(\mathbf{x}) d^3x.$$

4.3 Observables as operators; Hermitian operators; spectral decomposition

Operator. A (linear) operator $\hat{A}$ maps $\mathcal{H} \rightarrow \mathcal{H}$.

Adjoint. The adjoint $\hat{A}^\dagger$ is defined by

$$\langle \phi, \hat{A}\psi \rangle = \langle \hat{A}^\dagger \phi, \psi \rangle \quad \forall \phi, \psi \in \mathcal{H}.$$

Hermitian (self-adjoint) operator. $\hat{A}$ is Hermitian if $\hat{A}^\dagger = \hat{A}$. Hermitian operators have real expectation values:

$$\langle \psi, \hat{A}\psi \rangle \in \mathbb{R}.$$

Spectral decomposition (finite/discrete spectrum). If $\hat{A}$ has orthonormal eigenvectors $\{|a_n\rangle\}$ with eigenvalues $a_n \in \mathbb{R}$, then

$$\hat{A} = \sum_n a_n |a_n\rangle \langle a_n|, \quad \text{Id} = \sum_n |a_n\rangle \langle a_n|.$$

This expresses $\hat{A}$ as a weighted sum of projectors.

4.4 Projectors and measurement

Projector. A projector $\hat{P}$ satisfies $\hat{P}^2 = \hat{P}$ and (for an orthogonal projector) $\hat{P}^\dagger = \hat{P}$. The projector onto $|a\rangle$ is $\hat{P}_a = |a\rangle \langle a|$.

Measurement probability and state update. Given state $|\psi\rangle$, the probability of outcome associated with projector $\hat{P}$ is

$$\mathbb{P}(\text{outcome}) = \langle \psi, \hat{P}\psi \rangle.$$

If $\hat{P}$ occurs, the post-measurement state is $\hat{P}|\psi\rangle / \sqrt{\langle \psi, \hat{P}\psi \rangle}$.

4.5 Schrödinger equation, Hamiltonian, and potentials

Time-dependent Schrödinger equation. Let $\hbar > 0$ be Planck's constant divided by 2π . The Schrödinger equation is

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle,$$

where $\hat{H}$ is the Hamiltonian operator.

Generator interpretation. If $\hat{H}$ is time-independent, then

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(0)\rangle.$$

Thus $\hat{H}$ is the infinitesimal generator of *unitary* time evolution.

Single particle Hamiltonian with scalar potential. For mass $m > 0$ and potential energy $V(\mathbf{x}, t)$,

$$\hat{H} = -\frac{\hbar^2}{2m} \Delta + V(\mathbf{x}, t),$$

acting on $\psi(\mathbf{x}, t)$. The Laplacian $\Delta = \sum_{i=1}^3 \partial^2 / \partial x_i^2$ represents kinetic energy.

4.6 Gauge theory: connections, curvature, and why potentials matter

Global phase invariance. The physical state is unchanged by $|\psi\rangle \mapsto e^{i\alpha} |\psi\rangle$ for constant $\alpha \in \mathbb{R}$.

Local phase invariance and covariant derivative. If we demand invariance under $\alpha = \alpha(\mathbf{x}, t)$, then ordinary derivatives must be modified. Introduce a gauge potential A_μ (with components $A_0 = \phi$ and spatial $\mathbf{A}$) and define the covariant derivative

$$D_\mu := \partial_\mu + i\frac{q}{\hbar} A_\mu,$$

where q is electric charge and ∂_μ denotes spacetime derivatives.

Gauge transformation. Under $\psi \mapsto e^{iq\alpha/\hbar} \psi$, the potential must transform as

$$A_\mu \mapsto A_\mu - \partial_\mu \alpha$$

to keep $D_\mu \psi$ covariant.

Field strength as curvature. The electromagnetic field tensor is

$$F_{\mu\nu} := \partial_\mu A_\nu - \partial_\nu A_\mu.$$

In 3D vector calculus, $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{E} = -\nabla\phi - \partial_t \mathbf{A}$. Geometrically: A is a connection; F is its curvature [9, 10].

4.7 Abelian vs non-Abelian phases; degeneracy; Berry phase

Abelian phases. A $U(1)$ phase is a complex scalar $e^{i\varphi}$; phases commute: $e^{i\varphi_1}e^{i\varphi_2} = e^{i\varphi_2}e^{i\varphi_1}$. Electromagnetism is Abelian.

Non-Abelian phases. In non-Abelian gauge theories, “phases” are matrices $U \in SU(N)$ that generally do not commute. This requires path ordering in holonomies:

$$U(C) = \mathcal{P} \exp \left(\int_C A_\mu dx^\mu \right),$$

where $\mathcal{P}$ orders matrices along the path.

Degeneracy. An eigenvalue E of $\hat{H}$ is *degenerate* if there exist distinct orthonormal states $|\psi_a\rangle$ with $\hat{H}|\psi_a\rangle = E|\psi_a\rangle$. Degeneracy enables non-Abelian Berry phases (state mixing under adiabatic transport).

Berry phase (non-degenerate case). If $\hat{H}(\lambda)$ depends on parameters λ and the system is transported adiabatically around a loop in parameter space, a state can acquire a geometric phase γ [11, 12]:

$$\gamma = \oint \mathcal{A}(\lambda) \cdot d\lambda, \quad \mathcal{A}(\lambda) = i\langle\psi(\lambda) | \nabla_\lambda \psi(\lambda)\rangle.$$

This is holonomy in parameter space.

4.8 Magnetism, coils, and MRI: a physical instantiation of phase and Fourier decoding

Magnetic force and Lorentz law. A charged particle experiences

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}),$$

where $\mathbf{E}$ is electric field, $\mathbf{B}$ magnetic field, and $\mathbf{v}$ velocity.

Coil/solenoid field scale. For an ideal long solenoid with n turns per unit length and current I , the interior magnetic field is approximately

$$B \approx \mu_0 n I,$$

with μ_0 vacuum permeability.

MRI in one equation. In MRI, nuclear spins precess at Larmor frequency $\omega(\mathbf{r}) = \gamma B(\mathbf{r})$, where γ is the gyromagnetic ratio. Applying a gradient $B(\mathbf{r}) = B_0 + \mathbf{G} \cdot \mathbf{r}$ makes frequency position-dependent, so the measured signal becomes (schematically)

$$s(t) = \int \rho(\mathbf{r}) e^{i\gamma(\mathbf{G} \cdot \mathbf{r})t} d\mathbf{r},$$

which is a Fourier transform of the spin density $\rho(\mathbf{r})$. Reconstruction is an inverse Fourier transform. This is a concrete bridge: *phase accumulation (a gauge-like holonomy) encodes geometry; Fourier analysis decodes it* [22].

5 Part V: Stochastic Processes and Diffusion

5.1 Random variable vs stochastic process vs trajectories

Random variable. A map $X : \Omega \rightarrow \mathbb{R}^d$ is “one random outcome at one time”.

Stochastic process. A family $\{X_t\}_{t \geq 0}$ of random variables indexed by time t .

Trajectory/sample path. Fix $\omega \in \Omega$; then $t \mapsto X_t(\omega)$ is one realized path.

Ensemble/path measure. The distribution over all paths is a probability measure on a function space (path space). This distinction becomes central in Part VIII.

5.2 Brownian motion and the Wiener process

Definition (Wiener process). A (standard) Wiener process $\{W_t\}_{t \geq 0}$ is a real-valued stochastic process satisfying:

- $W_0 = 0$ almost surely,
- independent increments: $W_{t+s} - W_t$ independent of $\{W_u : u \leq t\}$,
- Gaussian increments: $W_{t+s} - W_t \sim \mathcal{N}(0, s)$,
- continuous sample paths almost surely.

Its variance grows linearly: $\text{Var}(W_t) = t$.

Scaling and non-differentiability. Typical increment size is $|W_{t+\Delta} - W_t| \sim \sqrt{\Delta}$. Hence “derivative” would scale like $\sqrt{\Delta}/\Delta = 1/\sqrt{\Delta} \rightarrow \infty$: Brownian paths are almost surely nowhere differentiable.

5.3 Itô integrals and Itô lemma

Why ordinary calculus breaks. For Brownian motion, the quadratic variation is

$$(W_{t+\Delta} - W_t)^2 \sim \Delta,$$

so formally $(dW_t)^2$ behaves like dt , not like $(dt)^2$. This makes second-order terms survive in limits.

Itô integral (definition by L^2 limit). Let X_t be an adapted process with $\mathbb{E}[\int_0^T X_t^2 dt] < \infty$. Define

$$\int_0^T X_t dW_t := \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} X_{t_k} (W_{t_{k+1}} - W_{t_k}),$$

where $0 = t_0 < t_1 < \dots < t_n = T$ and the mesh $\max_k (t_{k+1} - t_k) \rightarrow 0$. The key choice is evaluation at the *left endpoint* t_k (non-anticipating).

Itô isometry. A foundational identity:

$$\mathbb{E} \left[\left(\int_0^T X_t dW_t \right)^2 \right] = \mathbb{E} \left[\int_0^T X_t^2 dt \right].$$

This shows why completeness matters: the integral is defined as an L^2 limit in a complete space.

Itô SDE. An Itô stochastic differential equation (SDE) in $\mathbb{R}^d$ is

$$dX_t = b(X_t, t) dt + \sigma(X_t, t) dW_t,$$

where $b : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}^d$ is the drift, $\sigma : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}^{d \times m}$ is the diffusion matrix, and $W_t \in \mathbb{R}^m$ is an m -dimensional Wiener process.

Itô lemma (stochastic chain rule). Let $f : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}$ be twice differentiable in x and once in t . Then

$$df(X_t, t) = \partial_t f(X_t, t) dt + \nabla f(X_t, t)^\top dX_t + \frac{1}{2} \text{Tr} \left(\sigma \sigma^\top(X_t, t) \nabla^2 f(X_t, t) \right) dt,$$

where $\nabla^2 f$ is the Hessian matrix. The last term is the new piece caused by $(dW_t)^2 \sim dt$.

5.4 Master equation and Fokker–Planck equation

Master equation (discrete states). For a Markov jump process on discrete states $i \in \{1, 2, \dots\}$ with probabilities $P_i(t)$ and transition rates $W_{i \leftarrow j}$, probability conservation yields

$$\frac{dP_i(t)}{dt} = \sum_{j \neq i} \left(W_{i \leftarrow j} P_j(t) - W_{j \leftarrow i} P_i(t) \right).$$

This is the Kolmogorov forward equation for a continuous-time Markov chain.

Fokker–Planck equation (continuous state). For a diffusion process in $\mathbb{R}^d$ with density $p(x, t)$, the Fokker–Planck equation (FPE) is

$$\partial_t p(x, t) = - \sum_{i=1}^d \partial_i (b_i(x, t) p(x, t)) + \frac{1}{2} \sum_{i,j=1}^d \partial_i \partial_j (a_{ij}(x, t) p(x, t)),$$

where b is the drift and $a = \sigma \sigma^\top$ is the diffusion covariance [1, 2, 3].

Continuity form and probability current. Write

$$\partial_t p + \nabla \cdot J = 0,$$

with current

$$J(x, t) = b(x, t) p(x, t) - \frac{1}{2} \nabla \cdot (a(x, t) p(x, t)),$$

where $\nabla \cdot$ applied to a matrix means row-wise divergence producing a vector.

5.5 Langevin dynamics and fluctuation–dissipation

Underdamped Langevin equation. For position $x_t \in \mathbb{R}^d$ and velocity $v_t \in \mathbb{R}^d$:

$$m dv_t = -\gamma v_t dt - \nabla U(x_t) dt + \sqrt{2\gamma k_B T} dW_t, \quad dx_t = v_t dt,$$

where m is mass, γ friction, U potential energy, T temperature.

Fluctuation–dissipation theorem (FDT). The noise strength $\sqrt{2\gamma k_B T}$ is not arbitrary: it ensures that equilibrium is the Gibbs distribution $\propto e^{-\beta H}$ with $\beta = (k_B T)^{-1}$ [8, ?]. FDT states, structurally, that dissipation (friction) and fluctuations (noise) are linked by equilibrium consistency.

Overdamped Langevin (high friction). If inertia is negligible, one obtains

$$dx_t = -\mu \nabla U(x_t) dt + \sqrt{2D} dW_t,$$

where mobility $\mu = 1/\gamma$ and diffusion coefficient $D = \mu k_B T$ (Einstein relation).

5.6 Markov processes, semigroups, propagators, and infinitesimal generators

Markov property. A process $\{X_t\}$ is Markov if for all $s, t \geq 0$ and measurable sets A ,

$$\mathbb{P}(X_{t+s} \in A \mid X_u, 0 \leq u \leq t) = \mathbb{P}(X_{t+s} \in A \mid X_t).$$

Propagator (transition kernel). For a Markov process, define the transition density (or kernel) $p(t, x \mid x_0)$ such that

$$\mathbb{P}(X_t \in dx \mid X_0 = x_0) = p(t, x \mid x_0) dx.$$

It propagates densities:

$$p_t(x) = \int_{\mathbb{R}^d} p(t, x \mid x_0) p_0(x_0) dx_0.$$

Markov semigroup. Define operators $(P_t)_{t \geq 0}$ acting on functions f by

$$(P_t f)(x) := \mathbb{E}[f(X_t) \mid X_0 = x].$$

Then (P_t) satisfies the semigroup property:

$$P_{t+s} = P_t P_s, \quad P_0 = \text{Id}.$$

It is typically not invertible for $t > 0$ (irreversibility), hence a *semigroup* rather than a group.

Infinitesimal generator. The generator L is the derivative at $t = 0$:

$$(Lf)(x) := \lim_{t \downarrow 0} \frac{(P_t f)(x) - f(x)}{t},$$

for functions f in the domain of L .

Generator for Itô diffusions. For $dX_t = b(X_t)dt + \sigma(X_t)dW_t$ with $a = \sigma\sigma^\top$,

$$(Lf)(x) = b(x) \cdot \nabla f(x) + \frac{1}{2} \text{Tr}(a(x) \nabla^2 f(x)).$$

The Fokker–Planck equation is governed by the adjoint $L^\star$ acting on densities:

$$\partial_t p = L^\star p.$$

5.7 Fourier and Laplace transforms; characteristic functions; Green’s functions

Fourier transform. For $f : \mathbb{R} \rightarrow \mathbb{C}$, define

$$\widehat{f}(\omega) := \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt, \quad f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{f}(\omega) e^{i\omega t} d\omega.$$

Differentiation becomes multiplication: $\widehat{f'}(\omega) = i\omega \widehat{f}(\omega)$.

Laplace transform. For $t \geq 0$ and complex s with $\Re(s)$ large enough:

$$\mathcal{L}\{f\}(s) := \int_0^{\infty} e^{-st} f(t) dt.$$

It is a Fourier transform with exponential damping, suited for causal dynamics.

Characteristic function. For a real random variable X with density p , its characteristic function is

$$\varphi_X(k) := \mathbb{E}[e^{ikX}] = \int_{-\infty}^{\infty} e^{ikx} p(x) dx.$$

Thus φ_X is the Fourier transform of p . Moments are derivatives at $k = 0$ when they exist.

Green’s function. For a linear differential operator $\mathcal{L}$, a Green’s function G satisfies

$$\mathcal{L}G(\cdot, y) = \delta(\cdot - y),$$

so that $\mathcal{L}u = f$ has solution $u(x) = \int G(x, y) f(y) dy$. Propagators are Green’s functions of evolution equations.

6 Part VI: Information Geometry

6.1 Likelihood, log-likelihood, and the score

Likelihood. Given data x and parameter $\theta \in \mathbb{R}^n$, a model specifies a density $p(x \mid \theta)$. The likelihood is $L(\theta; x) = p(x \mid \theta)$.

Log-likelihood. Define $\ell(\theta; x) := \log p(x \mid \theta)$. The log form turns products into sums and makes derivatives meaningful.

Score function (parametric). The score is the gradient of log-likelihood:

$$s(\theta; x) := \nabla_\theta \log p(x \mid \theta) \in \mathbb{R}^n.$$

Under regularity conditions, $\mathbb{E}_\theta[s(\theta; X)] = 0$ when $X \sim p(\cdot \mid \theta)$.

6.2 Fisher information: sensitivity as curvature

Definition. The Fisher information matrix is

$$I(\theta) := \mathbb{E}_\theta \left[s(\theta; X) s(\theta; X)^\top \right].$$

Thus Fisher information is the covariance of the score:

$$I(\theta) = \text{Cov}_\theta(s(\theta; X)).$$

Local KL expansion (why Fisher is a metric). For nearby parameters θ and $\theta + d\theta$,

$$D_{\text{KL}}(p(\cdot | \theta) \| p(\cdot | \theta + d\theta)) = \frac{1}{2} d\theta^\top I(\theta) d\theta + o(\|d\theta\|^2).$$

Hence Fisher information is the local quadratic form (curvature) of KL divergence [6].

Dimensional interpretation. If θ has physical units, $I(\theta)$ has units of θ^{-2} : it measures how fast log-probability changes per unit change in θ .

6.3 Entropy, cross-entropy, KL divergence: one algebraic family

Entropy. For density p on $\mathbb{R}^d$,

$$H(p) := - \int_{\mathbb{R}^d} p(x) \log p(x) \, dx.$$

Cross-entropy. For densities p (truth) and q (model),

$$H(p, q) := - \int_{\mathbb{R}^d} p(x) \log q(x) \, dx.$$

KL divergence.

$$D_{\text{KL}}(p \| q) := \int_{\mathbb{R}^d} p(x) \log \frac{p(x)}{q(x)} \, dx.$$

Identity:

$$D_{\text{KL}}(p \| q) = H(p, q) - H(p) \geq 0,$$

with equality iff $p = q$ a.e.

6.4 Score as a vector field on probability space

Nonparametric score. For a density p on $\mathbb{R}^d$, define the *score field*

$$s_p(x) := \nabla_x \log p(x) \in \mathbb{R}^d.$$

This is a vector field on state space that depends on the distribution.

Why it appears in physics. In equilibrium statistical mechanics with density $p(x) \propto e^{-\beta U(x)}$,

$$\log p(x) = -\beta U(x) - \log Z, \quad \Rightarrow \quad s_p(x) = -\beta \nabla U(x).$$

Thus the score is proportional to (minus) the physical force. This is the clean conceptual bridge: *the score is a force-like object.*

Thermodynamic response and Fisher. In exponential families $p(x \mid \theta) \propto \exp(\theta^\top T(x) - A(\theta))$, Fisher information equals the Hessian of $A(\theta)$ and encodes susceptibilities (fluctuation–response) [6, 7].

7 Part VII: Diffusion Models (Core Target)

7.1 Forward diffusion as a controlled destruction of information

Discrete-time DDPM (high-level). Starting from data $x_0 \sim p_0$, define a Markov chain $x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_T$ that gradually adds Gaussian noise so that x_T is nearly Gaussian. Sampling is done by learning to reverse this chain [14, 15].

Continuous-time formulation (score-based SDEs). A forward SDE on $\mathbb{R}^d$ is

$$dx_t = f(x_t, t) dt + g(t) dW_t,$$

where $t \in [0, 1]$ is *diffusion time* (noise level), f is drift and g diffusion scale. This defines a family of marginals p_t satisfying an FPE. Score-based models learn $s_t(x) := \nabla_x \log p_t(x)$ and use time reversal to sample [16].

7.2 Ornstein–Uhlenbeck as the archetypal forward process

OU SDE. The OU process in $\mathbb{R}^d$:

$$dx_t = -\lambda x_t dt + \sqrt{2\lambda} dW_t,$$

with $\lambda > 0$. It is Gaussian and mean-reverting.

Stationary distribution. The stationary density is standard normal $p_\infty(x) \propto \exp(-\|x\|^2/2)$, and the score is $s_\infty(x) = -x$.

Why OU is central in diffusion models. OU is a linear Gaussian semigroup: it makes explicit the equivalence between (i) forward-noise semigroup, (ii) generator, (iii) exact scores. It is the simplest “laboratory” for score propagation.

7.3 Why the score appears in reverse-time dynamics

Time reversal principle. For a diffusion with forward drift f and diffusion scale g , the reverse-time SDE has drift involving the score of the forward marginals [13, 16]:

$$dx_t = \left(f(x_t, t) - g(t)^2 \nabla_x \log p_t(x_t) \right) dt + g(t) d\bar{W}_t,$$

where $\bar{W}_t$ is a Wiener process in reverse time. The key structural term is $-g(t)^2 s_t(x)$.

Interpretation as a force. The reverse drift contains a *force toward high-density regions* of p_t . Quantitatively: s_t has units of $1/(\text{units of } x)$; multiplying by g^2 (units x^2/t) yields x/t , a drift scale.

7.4 Score matching and Bayes-optimal denoising (Tweedie’s formula)

Gaussian corruption model. Let $x_0 \sim p_0$ and observe $x = x_0 + \sigma\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$ independent, $\sigma > 0$. Let p_σ be the density of x .

Denoiser. The Bayes-optimal denoiser is $\hat{x}(x) = \mathbb{E}[x_0 \mid x]$.

Tweedie’s identity. Under mild conditions,

$$\mathbb{E}[x_0 \mid x] = x + \sigma^2 \nabla_x \log p_\sigma(x).$$

Thus knowing the score gives the optimal denoiser. This is why diffusion models trained to predict scores also learn denoising [17, 18, 16].

7.5 Continuous vs discrete diffusion

Discrete diffusion. A finite number of steps with prescribed variance schedule; reverse transition kernels approximated by a neural net.

Continuous diffusion. An SDE defines a continuous semigroup; reverse-time sampling uses either (i) reverse SDE discretization, or (ii) probability flow ODE.

Why it matters. The research question in Part IX is naturally expressed in continuous time using semigroups and generators, and then approximated discretely for learning.

7.6 Connection to Schrödinger bridges and optimal transport

Schrödinger bridge (concept). Given an initial density and a final density, a Schrödinger bridge finds the most likely stochastic process connecting them relative to a reference diffusion, by minimizing path-space KL divergence [19, 20, 21]. This is an entropic regularization of optimal transport.

Diffusion models as bridges. Many diffusion-model constructions can be interpreted as learning controlled diffusions (or approximate bridges) that transform simple noise into data, with score fields encoding the control [16, 23].

8 Part VIII: Dynamic Objects and Two-Time Structure

8.1 Two notions of time and why conflating them causes conceptual fog

Internal (physical) time. Denote internal time by $\tau \in [0, T]$. It indexes the evolution of a physical system (frames of a video, states of a dynamical system).

Diffusion time. Denote diffusion time by $t \in [0, 1]$. It indexes the noise level in a diffusion model.

Two-time state. A dynamic object is a function $\tau \mapsto x^\tau \in \mathbb{R}^d$. A diffusion model builds noisy versions x_t^τ for each t :

$$x^\tau \longrightarrow x_t^\tau, \quad t \in [0, 1], \tau \in [0, T].$$

Thus (t, τ) label a *two-parameter family* of random variables.

8.2 Trajectories versus static datapoints: what changes mathematically

Static case. A dataset is samples $x \in \mathbb{R}^d$ with density $p(x)$. Scores live on $\mathbb{R}^d$:

$$s(x) = \nabla_x \log p(x).$$

Dynamic case. A dataset is trajectories $\mathbf{x} = (x^{\tau_1}, \dots, x^{\tau_N}) \in (\mathbb{R}^d)^N$ (discrete times) or functions $\tau \mapsto x^\tau$ (continuous). The probability object is a *path-space measure* $\mathbb{P}$ on trajectories. Scores may be:

- *per-frame (marginal) scores*: $s^\tau(x) = \nabla_x \log p^\tau(x)$ where p^τ is the marginal at time τ ,
- *trajectory-level score*: $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ in discrete time, or a functional derivative in continuous time.

Why this matters. Per-frame scores ignore temporal correlations; trajectory scores capture them but are high-dimensional. Learning how scores *propagate* in τ is a way to compress temporal structure.

8.3 Path-space probability measures and their scores

Discrete-time path distribution. For $\mathbf{x} = (x_1, \dots, x_N)$ with joint density $p(\mathbf{x})$, define the trajectory score

$$s(\mathbf{x}) := \nabla_{\mathbf{x}} \log p(\mathbf{x}) \in \mathbb{R}^{Nd}.$$

Its block components are $s_i(\mathbf{x}) = \nabla_{x_i} \log p(\mathbf{x})$, which generally depend on all times.

Markov factorization. If the physical dynamics is Markov in internal time,

$$p(x_1, \dots, x_N) = p(x_1) \prod_{i=1}^{N-1} p(x_{i+1} \mid x_i).$$

Then

$$\nabla_{x_i} \log p(\mathbf{x}) = \nabla_{x_i} \log p(x_i \mid x_{i-1}) + \nabla_{x_i} \log p(x_{i+1} \mid x_i)$$

(for interior indices), exposing local temporal coupling.

8.4 Per-frame vs trajectory-level scores under diffusion

Forward diffusion on trajectories. One may define a forward diffusion in the *trajectory space* $(\mathbb{R}^d)^N$:

$$d\mathbf{x}_t = f(\mathbf{x}_t, t) dt + g(t) d\mathbf{W}_t,$$

where $\mathbf{W}_t$ is an Nd -dimensional Wiener process. This yields a family of trajectory densities $p_t(\mathbf{x})$ and scores $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$.

Framewise diffusion approximation. Alternatively, one adds noise independently per frame given t , which makes the forward process factorize across τ but the *data distribution* remains temporally correlated. This is typical in practice and motivates learning architectures that respect both time axes [24].

9 Part IX: Central Research Question — Learning a Propagator for Scores Across Internal Time

9.1 What it means for a score to evolve across internal time

Statement. Fix a diffusion time t (a noise level). Suppose at each internal time τ we have a marginal density $p_t^\tau(x)$ on $\mathbb{R}^d$. Define the per-time score

$$s_t^\tau(x) := \nabla_x \log p_t^\tau(x).$$

We ask: is there an operator $\mathcal{K}_{\Delta\tau}$ such that

$$s_t^{\tau+\Delta\tau} \approx (\mathcal{K}_{\Delta\tau} s_t^\tau) \quad (\text{as functions of } x),$$

and what is the structure of $\mathcal{K}_{\Delta\tau}$?

Key conceptual point. Densities propagate *linearly* under Markov evolution (transfer operators), but scores involve log and ∇ , so their induced evolution is generally *nonlinear*. Understanding when and how this induced evolution becomes tractable (or learnable) is the core.

9.2 Why one should expect a propagator: semigroups in internal time

Physical internal-time dynamics. Assume the physical system has Markov evolution in internal time τ with semigroup $(\mathcal{P}_\tau)_{\tau \geq 0}$ acting on densities:

$$p^\tau = \mathcal{P}_\tau p^0, \quad \mathcal{P}_{\tau+s} = \mathcal{P}_\tau \mathcal{P}_s.$$

Equivalently, for observables f there is a Koopman semigroup $(\mathcal{U}_\tau)$:

$$(\mathcal{U}_\tau f)(x) = \mathbb{E}[f(X_\tau) \mid X_0 = x].$$

These are dual:

$$\int f(x)(\mathcal{P}_\tau p)(x) \, dx = \int (\mathcal{U}_\tau f)(x)p(x) \, dx.$$

This operator-theoretic structure is standard in dynamical systems and stochastic processes [5, 4].

Generator. There exists an internal-time generator $\mathcal{L}_\tau$ such that $\mathcal{P}_\tau = \exp(\tau \mathcal{L}^*)$ and $\mathcal{U}_\tau = \exp(\tau \mathcal{L})$ (time-homogeneous case). Learning *any* propagator implicitly learns (or approximates) a generator.

9.3 Explicit score propagator for deterministic invertible dynamics

Deterministic flow. Let internal dynamics be the ODE

$$\frac{d}{d\tau} x_\tau = b(x_\tau), \quad x_0 \sim p_0,$$

with flow map $\Phi_\tau : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that $x_\tau = \Phi_\tau(x_0)$. Assume Φ_τ is a diffeomorphism.

Density pushforward. The density at time τ is

$$p_\tau(x) = p_0(\Phi_{-\tau}(x)) |\det D\Phi_{-\tau}(x)|,$$

where $D\Phi_{-\tau}(x)$ is the Jacobian matrix.

Score transformation law. Take logs and gradients:

$$\log p_\tau(x) = \log p_0(\Phi_{-\tau}(x)) + \log |\det D\Phi_{-\tau}(x)|.$$

Differentiate:

$$s_\tau(x) := \nabla_x \log p_\tau(x) = (D\Phi_{-\tau}(x))^\top s_0(\Phi_{-\tau}(x)) + \nabla_x \log |\det D\Phi_{-\tau}(x)|.$$

This is an *explicit propagator* mapping s_0 to s_τ :

$$(\mathcal{K}_\tau s_0)(x) := (D\Phi_{-\tau}(x))^\top s_0(\Phi_{-\tau}(x)) + \nabla_x \log |\det D\Phi_{-\tau}(x)|.$$

It is generally nonlinear due to composition with $\Phi_{-\tau}$.

Limiting cases.

- If the flow is volume-preserving ($\det D\Phi_\tau \equiv 1$, e.g. Hamiltonian flows by Liouville's theorem), then the second term vanishes.
- If p_0 is uniform on a compact manifold and preserved by the flow, then $s_\tau \equiv 0$ for all τ .
- If Φ_τ is a rotation in $\mathbb{R}^d$ (orthogonal matrix), then $D\Phi_{-\tau}$ is that rotation matrix and the score rotates covariantly.

9.4 Score evolution PDE: generator-level description

Continuity equation. For deterministic drift b , density obeys the Liouville equation:

$$\partial_\tau p_\tau(x) = -\nabla \cdot (b(x)p_\tau(x)).$$

Let $u_\tau(x) := \log p_\tau(x)$ (log-density). Then

$$\partial_\tau u_\tau = \frac{\partial_\tau p_\tau}{p_\tau} = -\frac{\nabla \cdot (bp_\tau)}{p_\tau} = -b \cdot \nabla u_\tau - \nabla \cdot b.$$

Now define the score $s_\tau := \nabla u_\tau$. Differentiate:

$$\partial_\tau s_\tau = -\nabla(b \cdot s_\tau) - \nabla(\nabla \cdot b).$$

Expand $\nabla(b \cdot s)$ using product rule:

$$\nabla(b \cdot s) = (\nabla b)^\top s + (b \cdot \nabla)s,$$

where $(\nabla b)_{ij} = \partial_j b_i$ is the Jacobian matrix and $(b \cdot \nabla)s$ is directional derivative. Hence

$$\boxed{\partial_\tau s_\tau = -(\nabla b)^\top s_\tau - (b \cdot \nabla)s_\tau - \nabla(\nabla \cdot b).}$$

This is a linear first-order PDE for the score given the drift b .

Why this is propagation. The PDE is the infinitesimal form of the propagator $\mathcal{K}_\tau$ above. For small $\Delta\tau$,

$$s_{\tau+\Delta\tau} \approx s_\tau + \Delta\tau \mathcal{G}[s_\tau],$$

where $\mathcal{G}$ is the *score generator* defined by the right-hand side. Learning a propagator is learning (an approximation of) this generator.

9.5 Including stochastic physical dynamics: nonlinearity from diffusion

Physical internal diffusion. If internal-time dynamics is an SDE

$$dX_\tau = b(X_\tau)d\tau + \sqrt{2D}dW_\tau,$$

then p_τ solves

$$\partial_\tau p_\tau = -\nabla \cdot (bp_\tau) + D\Delta p_\tau.$$

For $u_\tau = \log p_\tau$,

$$\partial_\tau u_\tau = -b \cdot s_\tau - \nabla \cdot b + D \frac{\Delta p_\tau}{p_\tau}.$$

Using $\Delta p = p(\Delta u + \|\nabla u\|^2)$, we get

$$\partial_\tau u_\tau = -b \cdot s_\tau - \nabla \cdot b + D(\nabla \cdot s_\tau + \|s_\tau\|^2).$$

Taking gradients yields a score PDE with a *nonlinear* term $\nabla \|s_\tau\|^2$:

$$\partial_\tau s_\tau = -\nabla(b \cdot s_\tau) - \nabla(\nabla \cdot b) + D\nabla(\nabla \cdot s_\tau) + D\nabla \|s_\tau\|^2.$$

Thus, even if density propagation is linear, score propagation is typically nonlinear under stochastic physical dynamics.

9.6 Relation to Koopman and transfer operators: where the propagator lives

Koopman operator (observables). For deterministic flow $\Phi_{\Delta\tau}$, the Koopman operator acts on $f : \mathbb{R}^d \rightarrow \mathbb{R}$ by

$$(\mathcal{U}_{\Delta\tau}f)(x) := f(\Phi_{\Delta\tau}(x)).$$

Its generator is $(\mathcal{L}f)(x) = b(x) \cdot \nabla f(x)$.

Transfer/Perron–Frobenius operator (densities). The transfer operator $\mathcal{P}_{\Delta\tau}$ acts on densities so that

$$p_{\tau+\Delta\tau} = \mathcal{P}_{\Delta\tau}p_\tau.$$

For invertible flows, $\mathcal{P}_{\Delta\tau}$ is induced by the Jacobian formula. For Markov diffusions, $\mathcal{P}_{\Delta\tau} = \exp(\Delta\tau\mathcal{L}^*)$.

Score propagator as induced operator. The score map $p \mapsto \nabla \log p$ is nonlinear, so the score propagator is the composition

$$s_{\tau+\Delta\tau} = \nabla \log (\mathcal{P}_{\Delta\tau}p_\tau).$$

Learning a score propagator is, in principle, learning the action of $\mathcal{P}_{\Delta\tau}$ *modulo* the nonlinear map $\nabla \log$.

9.7 When the propagator reduces to simple transformations

Rotations (orthogonal dynamics). If $\Phi_{\Delta\tau}(x) = Rx$ with $R^\top R = I$, then $\det D\Phi = 1$ and

$$s_{\tau+\Delta\tau}(x) = R^\top s_\tau(R^\top x),$$

a pure covariant rotation.

Linear drift (Gaussian family closure). If p_τ stays Gaussian with mean μ_τ and covariance Σ_τ , then

$$s_\tau(x) = \nabla \log \mathcal{N}(x; \mu_\tau, \Sigma_\tau) = -\Sigma_\tau^{-1}(x - \mu_\tau),$$

a linear field. Propagating s_τ reduces to propagating (μ_τ, Σ_τ) under known equations. This is the simplest setting where score propagation is *finite-dimensional* and can be represented by matrix operations.

Pure diffusion on $\mathbb{R}^d$. If internal dynamics is heat flow, $p_\tau = p_0 * \mathcal{N}(0, 2D\tau I)$. Score fields evolve by a PDE that can be studied via Fourier transform; the propagator is linear on p but nonlinear on s . Understanding approximations here informs score propagation in diffusion models.

9.8 Two-time propagators: coupling internal time and diffusion time

Two-parameter family. In dynamic diffusion models, one can view the object as having a joint evolution:

$$(\tau, t) \mapsto p_t^\tau.$$

Internal dynamics (physics) propagates in τ ; diffusion forward/noise propagates in t .

Non-commutativity as a research-relevant possibility. Internal evolution and diffusion-time evolution may not commute:

$$\mathcal{P}_\tau^{\text{phys}} \circ \mathcal{P}_t^{\text{diff}} \neq \mathcal{P}_t^{\text{diff}} \circ \mathcal{P}_\tau^{\text{phys}},$$

especially when diffusion is applied in a representation space tied to τ -dependent structure (e.g. constraints, symmetries). This suggests viewing the pair as a *two-dimensional connection* (echoing gauge theory language) or as a non-commuting semigroup action. The board notes on score evolution across internal time emphasize this two-time structure and its operator consequences [25].

9.9 Learning a score propagator $\leftrightarrow$ learning physical dynamics

Core identifiability idea. For deterministic dynamics, the score propagator depends on $\Phi_{-\tau}$ and its Jacobian. Therefore, if one can learn $\mathcal{K}_\tau$ accurately over enough distributions, one constrains Φ_τ and hence the drift field b .

Generator recovery viewpoint. From the score PDE

$$\partial_\tau s_\tau = -(\nabla b)^\top s_\tau - (b \cdot \nabla) s_\tau - \nabla(\nabla \cdot b),$$

b enters in three places: advection $(b \cdot \nabla)s$, linear deformation $(\nabla b)^\top s$, and divergence forcing $\nabla(\nabla \cdot b)$. If s_τ is observed/estimated across many τ and across diverse initial distributions, one can pose an inverse problem for b .

Why diffusion-model scores help. Diffusion models provide access to scores s_t^τ at many noise levels t . Varying t changes smoothing (convolution with Gaussians in many setups), which can regularize and improve identifiability of derivatives and Jacobians. This is aligned with the physics-inspired viewpoint in M'ezard's updated generative diffusion notes [23].

Operator-learning viewpoint (Koopman/transfer). Learning $\mathcal{K}_{\Delta\tau}$ is analogous to learning Koopman operators for observables or transfer operators for densities. The crucial difference is that $\mathcal{K}$ acts on *scores*, i.e. on a derived geometric object (a log-density gradient). This suggests:

- **Linearization strategies:** embed scores into feature spaces where propagation is approximately linear (Koopman-inspired),
- **Generator learning:** learn an infinitesimal operator for score evolution, then exponentiate to a propagator,
- **Physics constraints:** enforce invariances (e.g. volume preservation $\Rightarrow \nabla \cdot b = 0$) that simplify $\mathcal{K}$.

9.10 Concrete formulation: score propagator as a learned semigroup on vector fields

Function space. Let $\mathcal{V}$ be a function space of vector fields on $\mathbb{R}^d$, e.g. $\mathcal{V} = L^2(p; \mathbb{R}^d)$ for some reference measure p or a Sobolev space capturing derivatives.

Propagator family. We seek $(\mathcal{K}_{\Delta\tau})_{\Delta\tau \geq 0}$ such that

$$\mathcal{K}_0 = \text{Id}, \quad \mathcal{K}_{\Delta\tau_1 + \Delta\tau_2} = \mathcal{K}_{\Delta\tau_1} \circ \mathcal{K}_{\Delta\tau_2},$$

at least approximately. This is a semigroup property in internal time. Its generator $\mathcal{G}$ satisfies

$$\mathcal{K}_{\Delta\tau} \approx \exp(\Delta\tau \mathcal{G}).$$

Why this is not tautological. Even if underlying densities are Markov, the induced action on scores is nonlinear and may not close on per-frame scores unless additional assumptions hold (Gaussian closure, weak-noise limits, local Markov structure). Establishing and exploiting approximate closure is exactly the research opportunity highlighted in Score-Based Generative Diffusion Models for Dynamic Objects internal notes [24].

9.11 Summary of the research agenda in one diagram

Let τ denote physical time and t diffusion time. Then:

$$\text{physical dynamics } (b, \sigma) \Rightarrow \text{transfer semigroup } \mathcal{P}_\tau \Rightarrow p^\tau \Rightarrow s^\tau = \nabla \log p^\tau.$$

Diffusion models give access to (p_t^τ, s_t^τ) across t . The research goal is to learn an operator $\mathcal{K}_{\Delta\tau}$ such that

$$s_t^{\tau + \Delta\tau} \approx \mathcal{K}_{\Delta\tau} [s_t^\tau],$$

and to relate $\mathcal{K}$ (or its generator) back to (b, σ) .

Interpretive punchline. Learning a propagator for the score across internal time is, in a precise operator-theoretic sense, a way of learning *how probability mass is transported by the underlying physics*. This is learning dynamics expressed in the geometry of log-densities rather than in raw states.

Acknowledgment of mandatory internal sources

The note conceptually integrates three internal sources as requested:

- M. M'ezard, *Generative diffusion: updated notes* (Dec. 2025) [23],
- *Score-Based Generative Diffusion Models for Dynamic Objects* (internal thesis notes) [24],
- *Board notes on score evolution across internal time* [25].

These are cited for conceptual alignment and research framing; the present note remains self-contained in definitions and derivations.

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